

ARP POISONING AND PERFORMING MAN IN THE MIDDLE ATTACK

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Introduction

ARP (Address Resolution Protocol)

- ARP (Address Resolution Protocol) maps IP addresses to MAC addresses in a local network.
- **Process:** When a device wants to communicate, it broadcasts an ARP request asking “Who has IP X?” The device with that IP responds with its MAC address.
- **ARP Cache:** Devices store these mappings temporarily in a table (arp -a shows this).
- **Weakness:** ARP has no authentication, so it trusts any reply it receives.

ARP Spoofing

- **Concept:** An attacker sends forged ARP replies, tricking a victim into associating the gateway’s IP with the attacker’s MAC.
- **Result:** Victim traffic meant for the gateway is rerouted through the attacker.
- **Impact:** Enables packet interception, modification, or blocking.

Machines

- Kali Linux - Attacker machine
- Windows VM - Victim’s machine

Man-in-the-Middle (MITM) Attack

A MITM attack occurs when an attacker secretly relays or alters communication between two parties who believe they are directly communicating.

- **In ARP spoofing:**

- Victim thinks it's talking to the gateway.
- Gateway thinks it's talking to the victim.
- Attacker sits in the middle, controlling traffic.

TOOL USED : ETTERCAP

Ettercap is a comprehensive network sniffer and packet analyzer that supports active and passive analysis of many protocols. Originally built to perform MITM attacks on LANs

Key Features

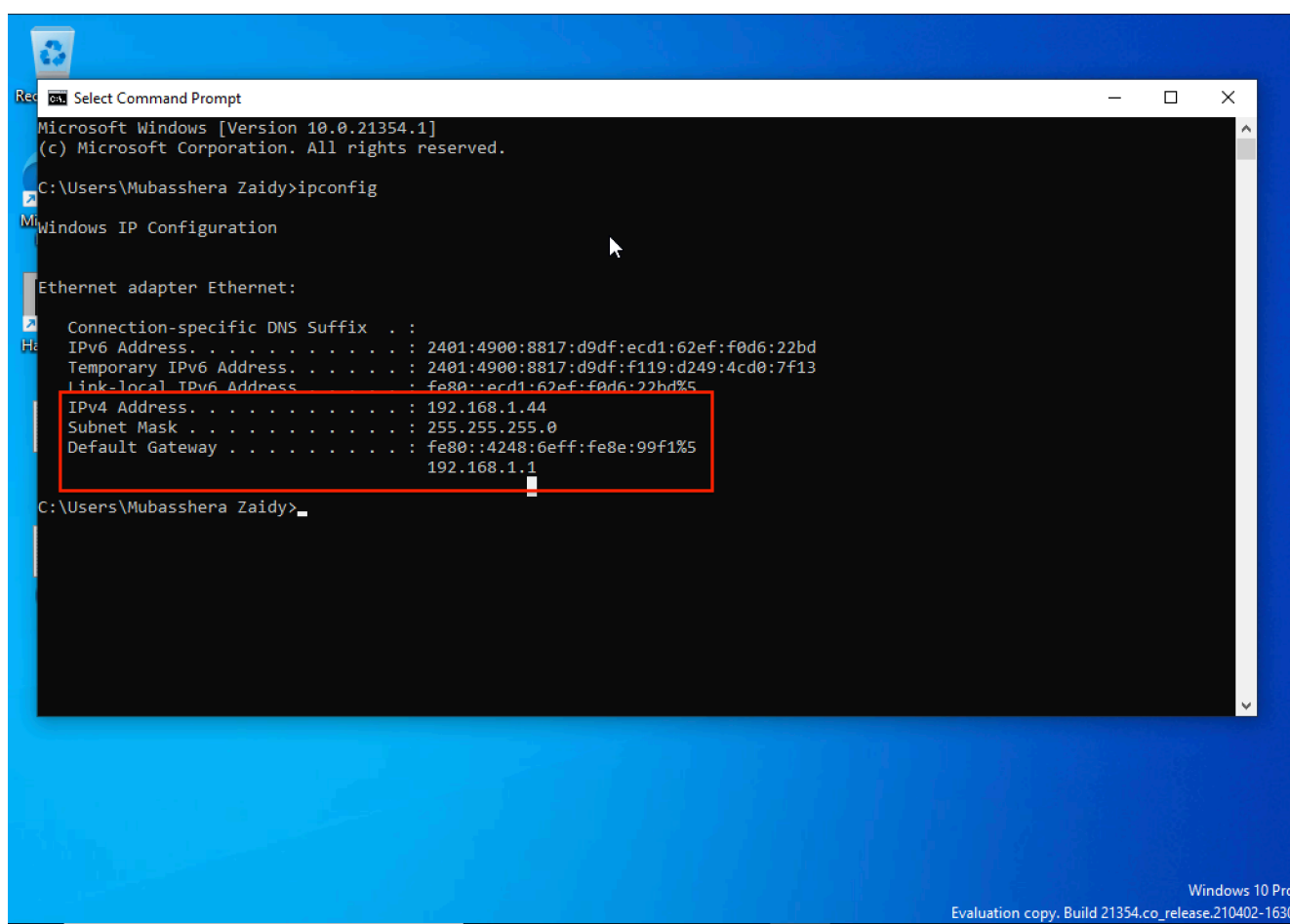
- **ARP Poisoning:** Core functionality to trick devices into sending traffic through the attacker's machine.
- **Traffic Interception:** Capture and analyze protocols like HTTP, FTP, and more.
- **Modification:** Inject, replace, or delete data in live connections.
- **Plugins:** Extend functionality with filters and scripts for advanced attacks.
- **Cross-Platform:** Works on Linux, macOS, and other Unix-like systems.

METHODOLOGY

STEP 1 : Identify the Victim's IP and Default Gateway

- Victim's Ip - 192.168.1.44
- Default Gateway : 192.168.1.1

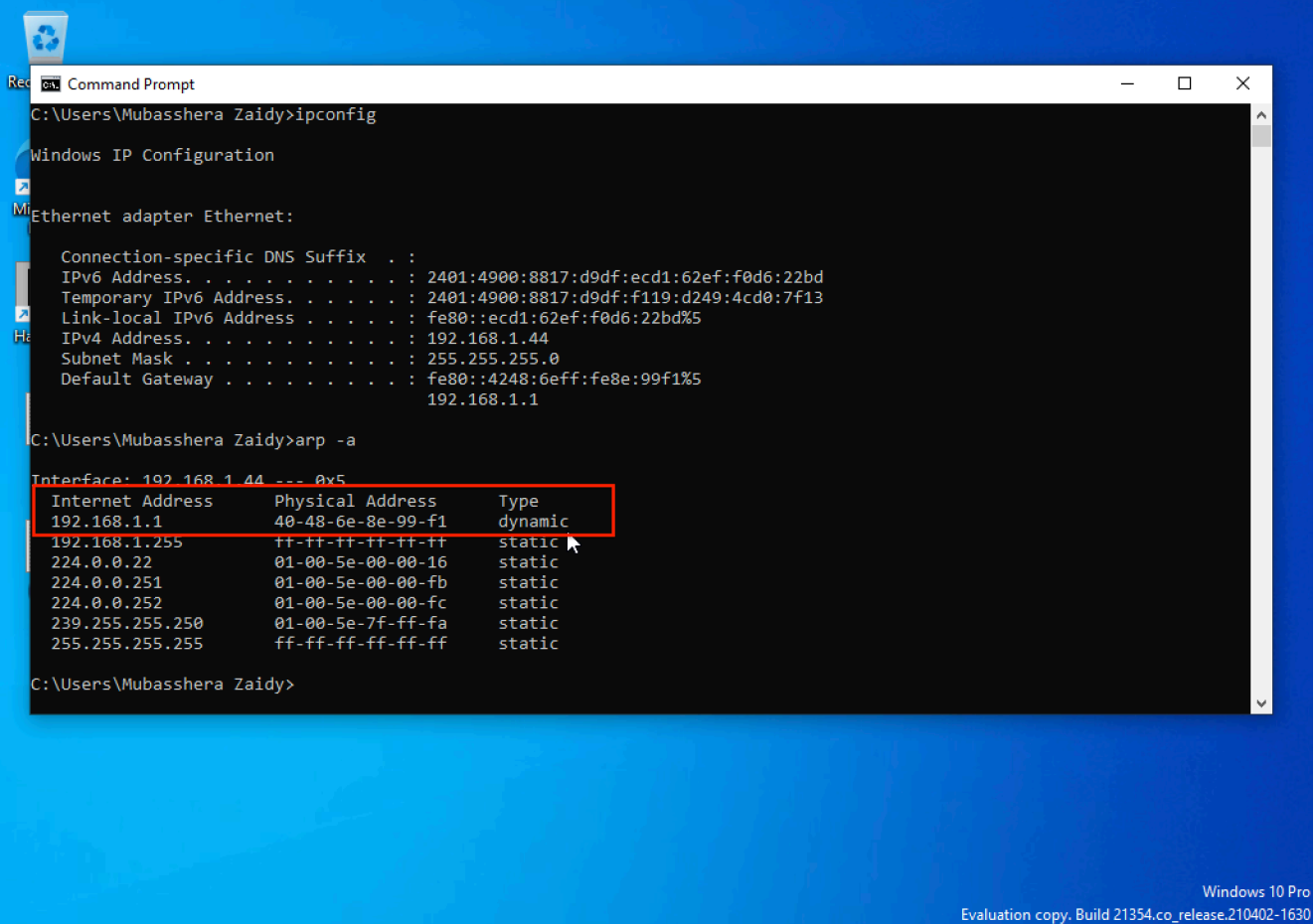
This means any packets sent or received by the victim are being directed through this gateway.



STEP 2 : Now type arp -a to view the arp table and identify the MAC address of the default gateway.

Default Gateway Ip address - 192.168.1.1

Default Gateway MAC address - 40-48-6e-8e-99-f1



```
C:\Users\Mubasshera Zaidy>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2401:4900:8817:d9df:ecd1:62ef:f0d6:22bd
    Temporary IPv6 Address. . . . . : 2401:4900:8817:d9df:f119:d249:4cd0:7f13
    Link-local IPv6 Address . . . . . : fe80::ecd1:62ef:f0d6:22bd%5
    IPv4 Address. . . . . : 192.168.1.44
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::4248:6eff:fe8e:99f1%5
                               192.168.1.1

C:\Users\Mubasshera Zaidy>arp -a

Interface: 192.168.1.44 --- 0x5

Internet Address      Physical Address      Type
192.168.1.1           40-48-6e-8e-99-f1    dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

C:\Users\Mubasshera Zaidy>
```

STEP 3 : Identify the attacker's IP and MAC address

- Attacker's IP address : 192.168.1.3
- MAC address : c2:30:bc:5e:5d:f6

```
(root@kali)~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.3 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::c030:bcff:fe5e:5df6 prefixlen 64 scopeid 0x20<link>
    inet6 2401:4900:8817:d9df:f9c1:5f9c:ed9a:867b prefixlen 64 scopeid 0x0<global>
    inet6 2401:4900:8817:d9df:c030:bcff:fe5e:5df6 prefixlen 64 scopeid 0x0<global>
    ether c2:30:bc:5e:5d:f6 txqueuelen 1000 (Ethernet)
    RX packets 49194 bytes 11906080 (11.3 MiB)
    RX errors 0 dropped 2141 overruns 0 frame 0
    TX packets 13597 bytes 1383062 (1.3 MiB)
    TX errors 0 dropped 11 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 897 bytes 54363 (53.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 897 bytes 54363 (53.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(root@kali)~#
```

STEP 4 : Enable IP forwarding by typing the following command `echo 1 > /proc/sys/net/ipv4/ip_forward`

To check if the forwarding is enabled or not type : `sysctl net.ipv4.ip_forward`. If it returns 1 then it means it is enabled.

This step is important because by default, when an ARP request or packet destined for another host arrives at our system, the kernel does not forward it. It simply rejects or drops the traffic because the machine is not acting as a router.

When we enable IP forwarding using the above command the system changes behaviour, instead of discarding those packets, it forwards them to the default gateway.

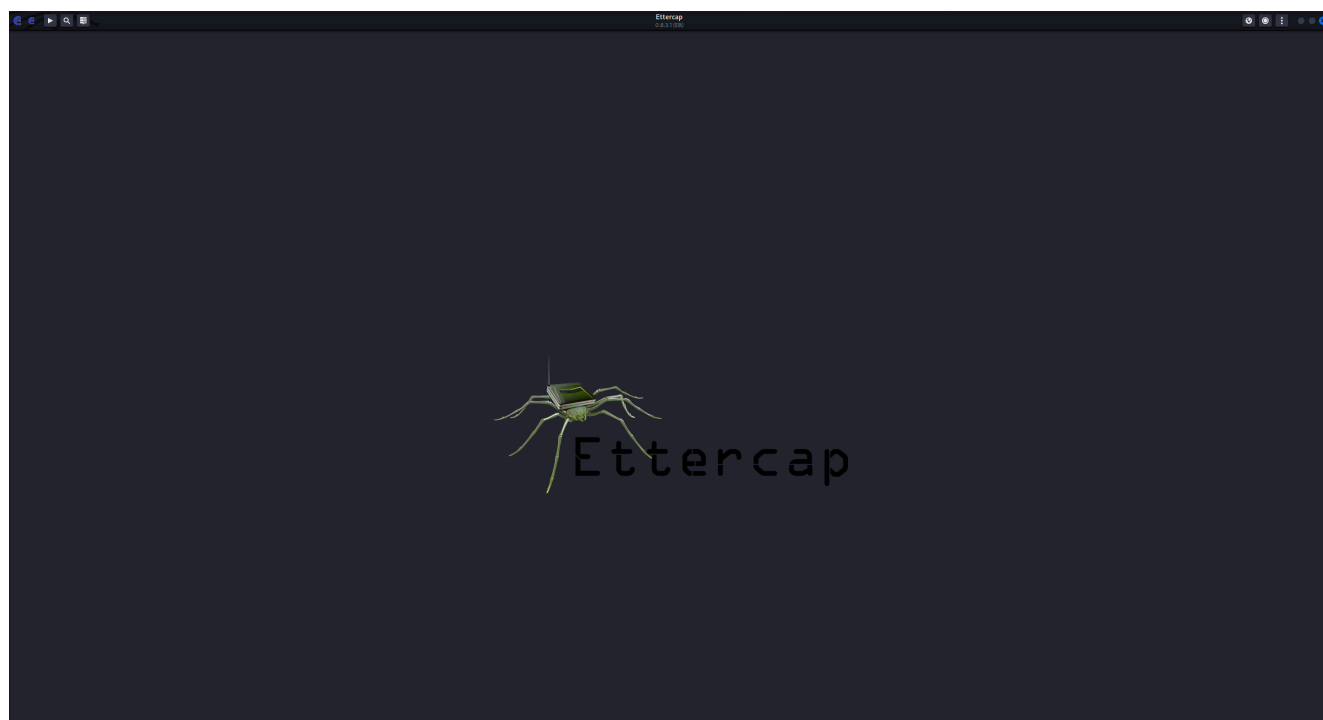
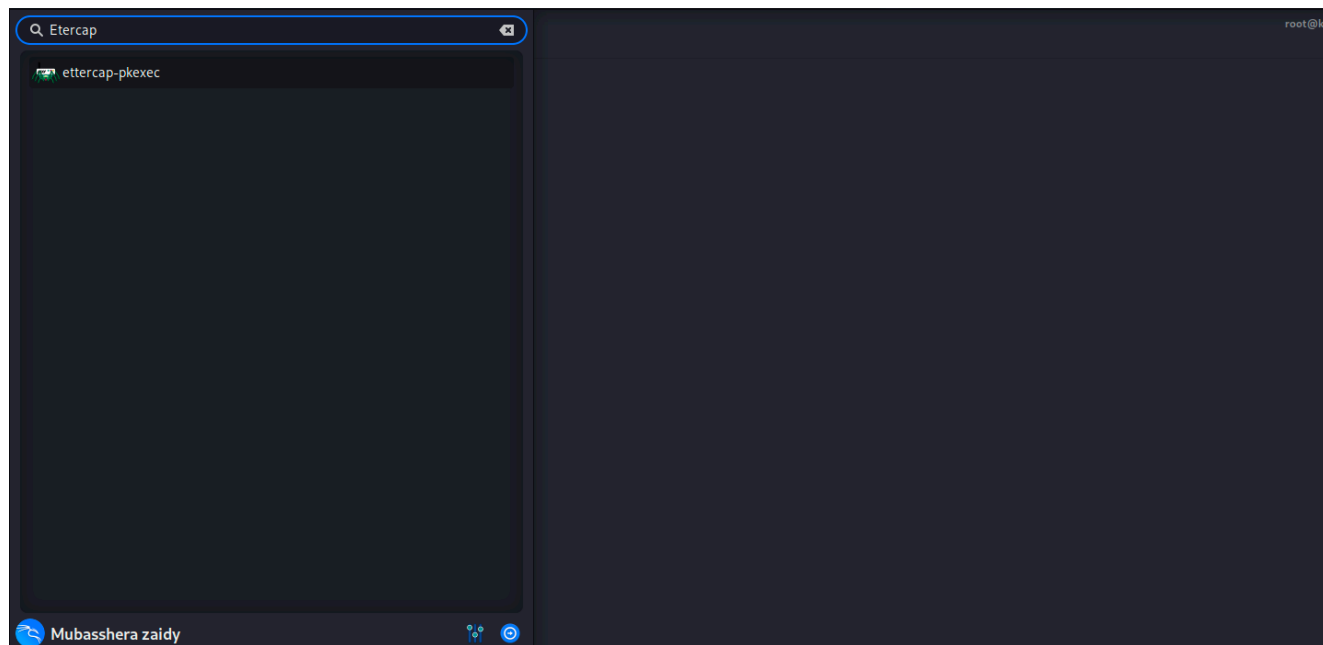
This ensures that even after ARP spoofing, the victim's traffic continues to reach the gateway, maintaining internet connectivity while still allowing us to intercept and analyze the packets.

```
(root@kali)-[/home/mubassherazaidy]
# echo 1 > /proc/sys/net/ipv4/ip_forward

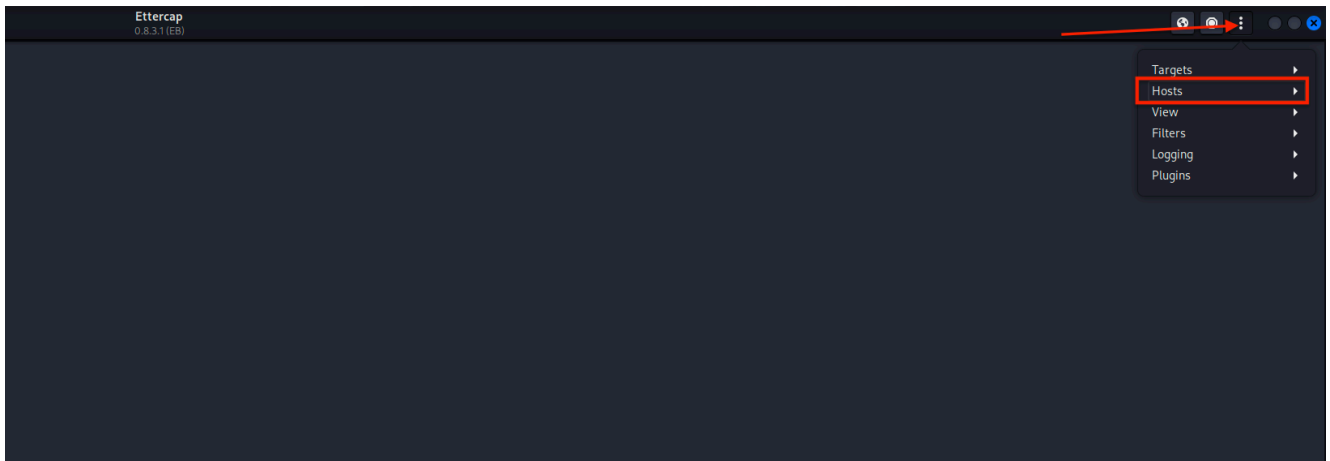
(root@kali)-[/home/mubassherazaidy]
# sysctl net.ipv4.ip_forward
net.ipv4.ip_forward = 1

(root@kali)-[/home/mubassherazaidy]
#
```

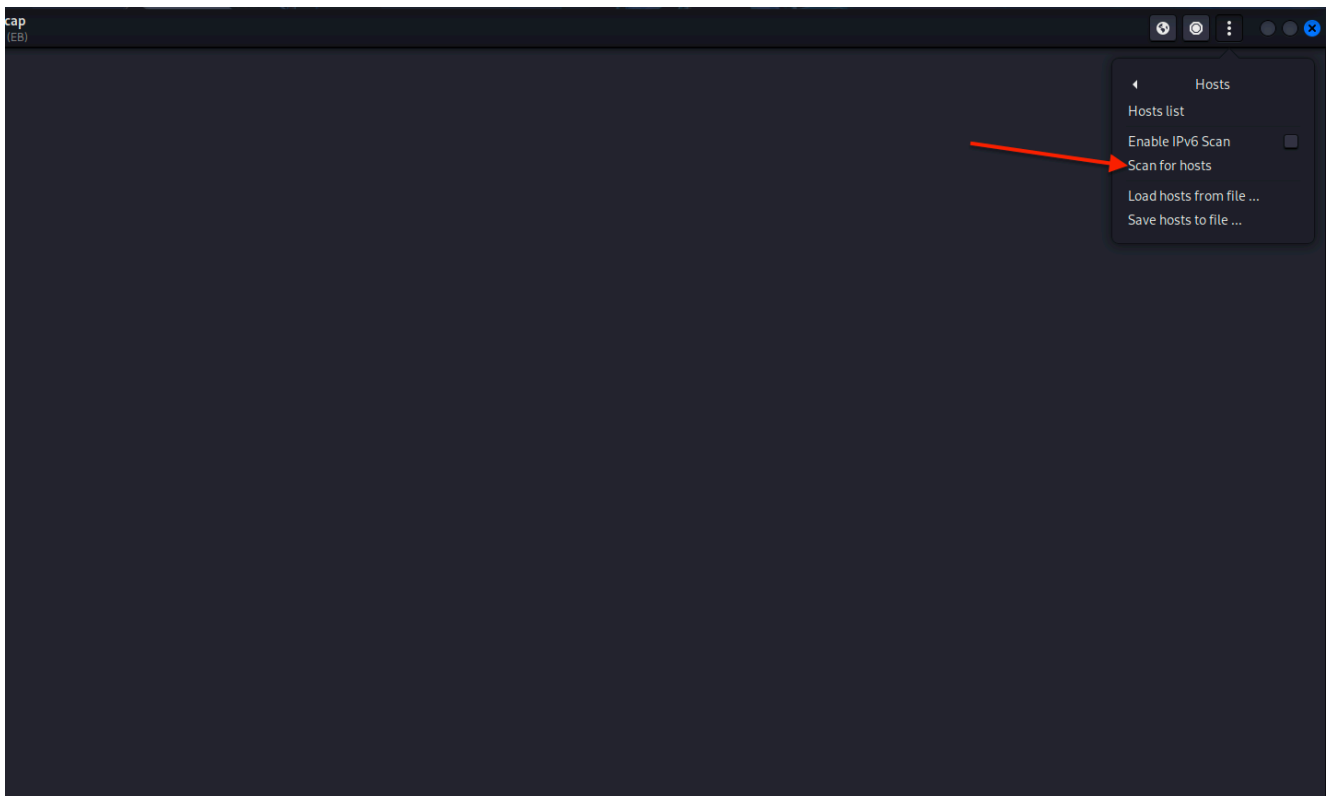
STEP 5: Now in kali, type Ettercap in search bar and click on ettercap-pkexec



STEP 6 : Once it is started click on : provided on right-top of the page and click on Hosts.



STEP 7 : Under Hosts click on select "Scan for hosts"



- It has found 9 hosts

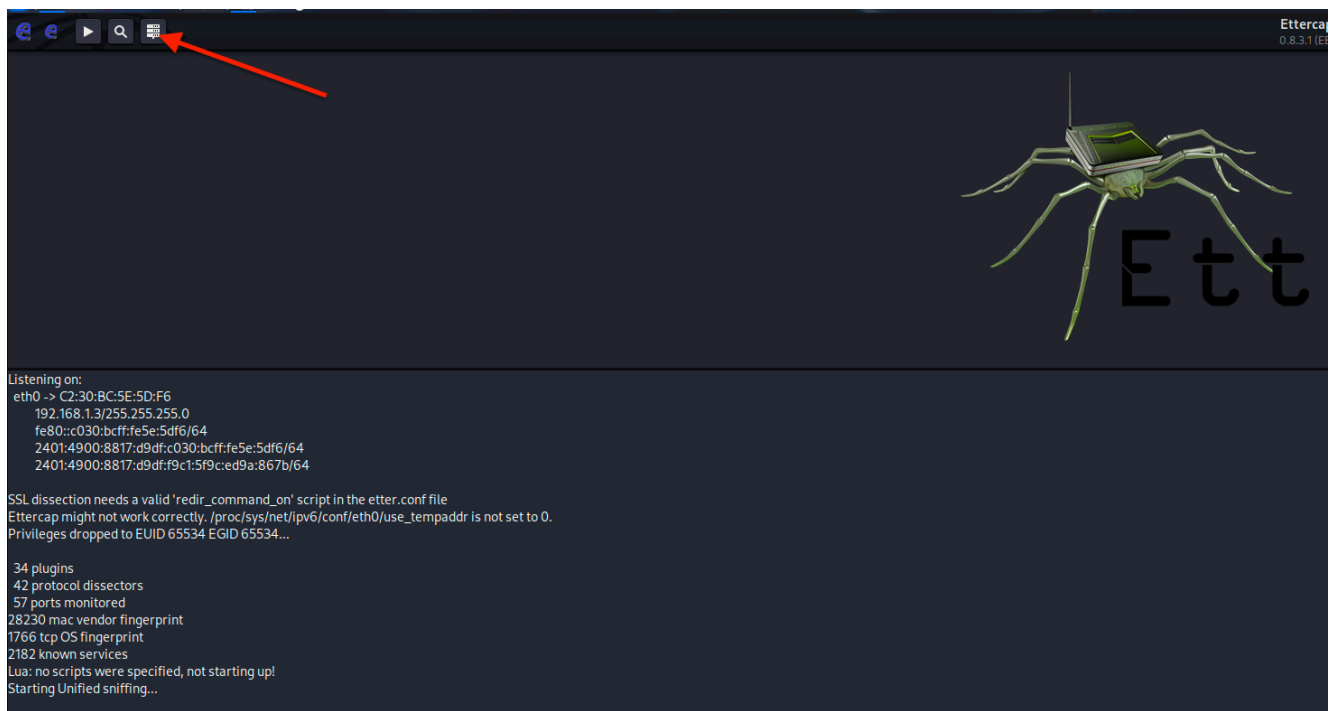
```
Listening on:
eth0 -> C2:30:BC:5E:5D:F6
192.168.1.3/255.255.255.0
fe80::c030:bcff:fe5e:5df6/64
2401:4900:8817:d9df:c030:bcff:fe5e:5df6/64
2401:4900:8817:d9df:d5fc:1c8d:7d24:89ea/64

SSL dissection needs a valid 'redir_command_on' script in the etter.conf file
Ettercap might not work correctly. /proc/sys/net/ipv6/conf/eth0/use_tempaddr is not set to 0.
Privileges dropped to EUID 65534 EGID 65534...

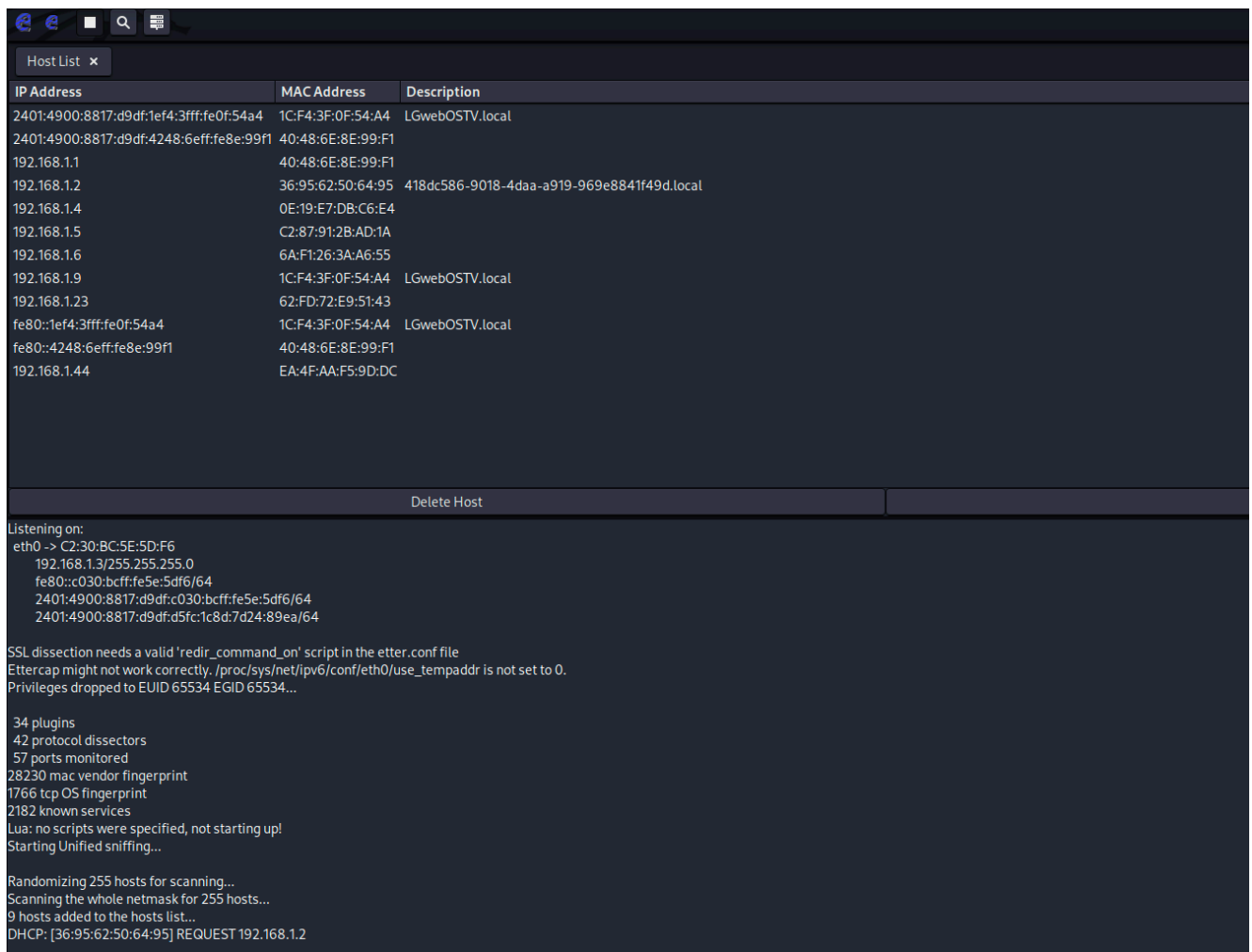
34 plugins
42 protocol dissectors
57 ports monitored
28230 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!
Starting Unified sniffing...

Randomizing 255 hosts for scanning...
Scanning the whole netmask for 255 hosts...
9 hosts added to the hosts list...
```

STEP 8 : Click on the lists icon to list the hosts



- It has listed all the hosts



The screenshot shows a terminal window with a dark background. At the top, there's a tab labeled "Host List" with a close button. Below the tab is a table with three columns: "IP Address", "MAC Address", and "Description". The table contains several entries, including IP addresses like 2401:4900:8817:d9df:1ef4:3fff:fe0f:54a4 and 192.168.1.1, and MAC addresses like 1C:F4:3F:0F:54:A4 and 40:48:6E:8E:99:F1. Some entries have descriptions like "LGwebOSTV.local" and "418dc586-9018-4daa-a919-969e8841f49d.local".

IP Address	MAC Address	Description
2401:4900:8817:d9df:1ef4:3fff:fe0f:54a4	1C:F4:3F:0F:54:A4	LGwebOSTV.local
2401:4900:8817:d9df:4248:6eff:fe8e:99f1	40:48:6E:8E:99:F1	
192.168.1.1	40:48:6E:8E:99:F1	
192.168.1.2	36:95:62:50:64:95	418dc586-9018-4daa-a919-969e8841f49d.local
192.168.1.4	0E:19:E7:DB:C6:E4	
192.168.1.5	C2:87:91:2B:AD:1A	
192.168.1.6	6A:F1:26:3A:A6:55	
192.168.1.9	1C:F4:3F:0F:54:A4	LGwebOSTV.local
192.168.1.23	62:FD:72:E9:51:43	
fe80::1ef4:3fff:fe0f:54a4	1C:F4:3F:0F:54:A4	LGwebOSTV.local
fe80::4248:6eff:fe8e:99f1	40:48:6E:8E:99:F1	
192.168.1.44	EA:4F:AA:F5:9D:DC	

Below the table, there's a "Delete Host" button. At the bottom of the terminal, there's a large block of text showing network scanning output:

```

Listening on:
eth0 -> C2:30:BC:5E:5D:F6
192.168.1.3/255.255.255.0
fe80::c030:bcff:fe5e:5df6/64
2401:4900:8817:d9df:c030:bcff:fe5e:5df6/64
2401:4900:8817:d9df:d5fc:1c8d:7d24:89ea/64

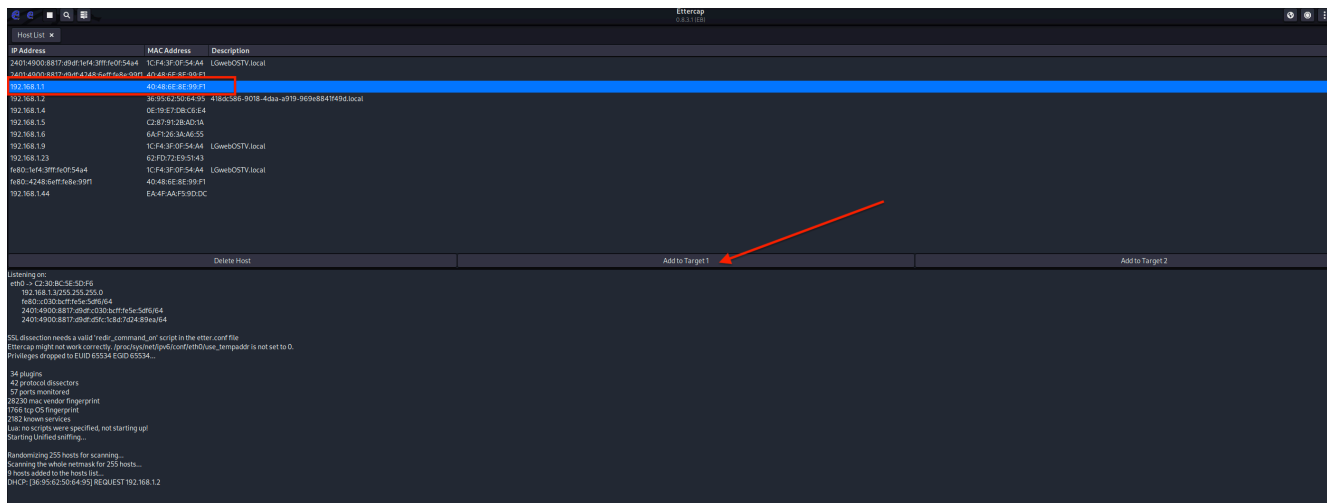
SSL dissection needs a valid 'redir_command_on' script in the etter.conf file
Ettercap might not work correctly. /proc/sys/net/ipv6/conf/eth0/use_tempaddr is not set to 0.
Privileges dropped to EUID 65534 EUID 65534...

34 plugins
42 protocol dissectors
57 ports monitored
28230 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!
Starting Unified sniffing...

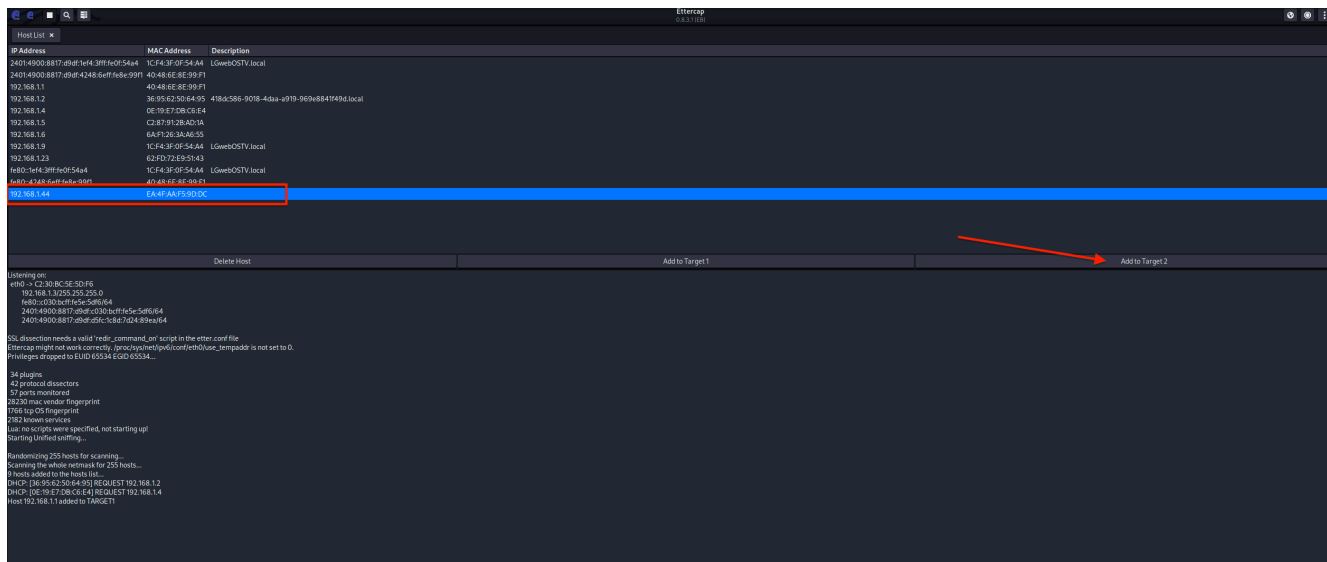
Randomizing 255 hosts for scanning...
Scanning the whole netmask for 255 hosts...
9 hosts added to the hosts list...
DHCP: [36:95:62:50:64:95] REQUEST 192.168.1.2

```

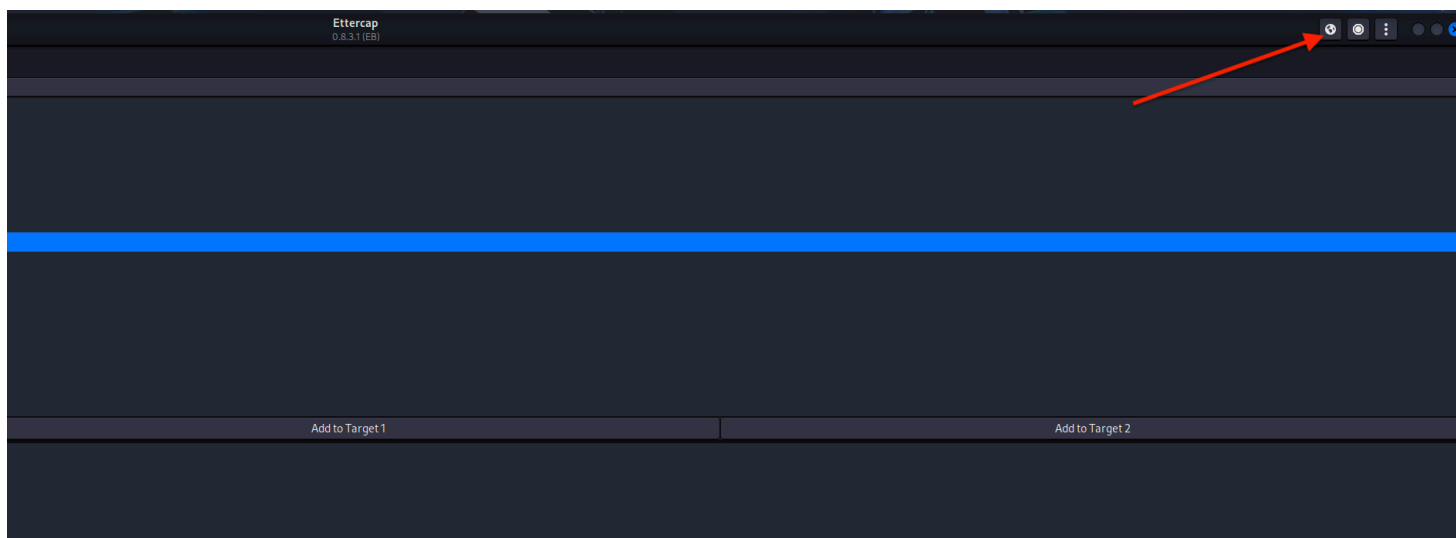
STEP 9 : Now select the 192.168.1.1 which is the default gateway IP address and click on Add to Target 1



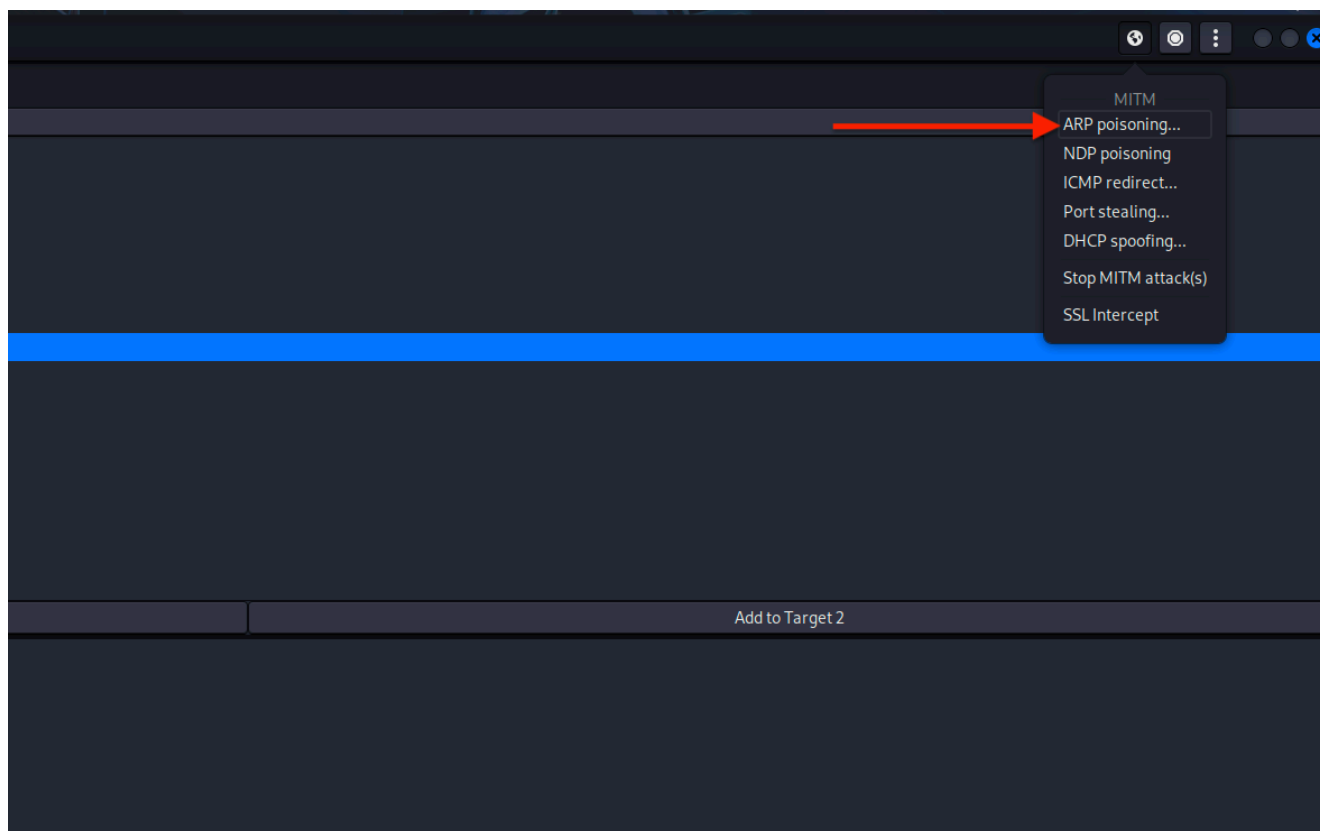
Select 192.168.1.44 which is the victim's IP address and click on Add to Target 2



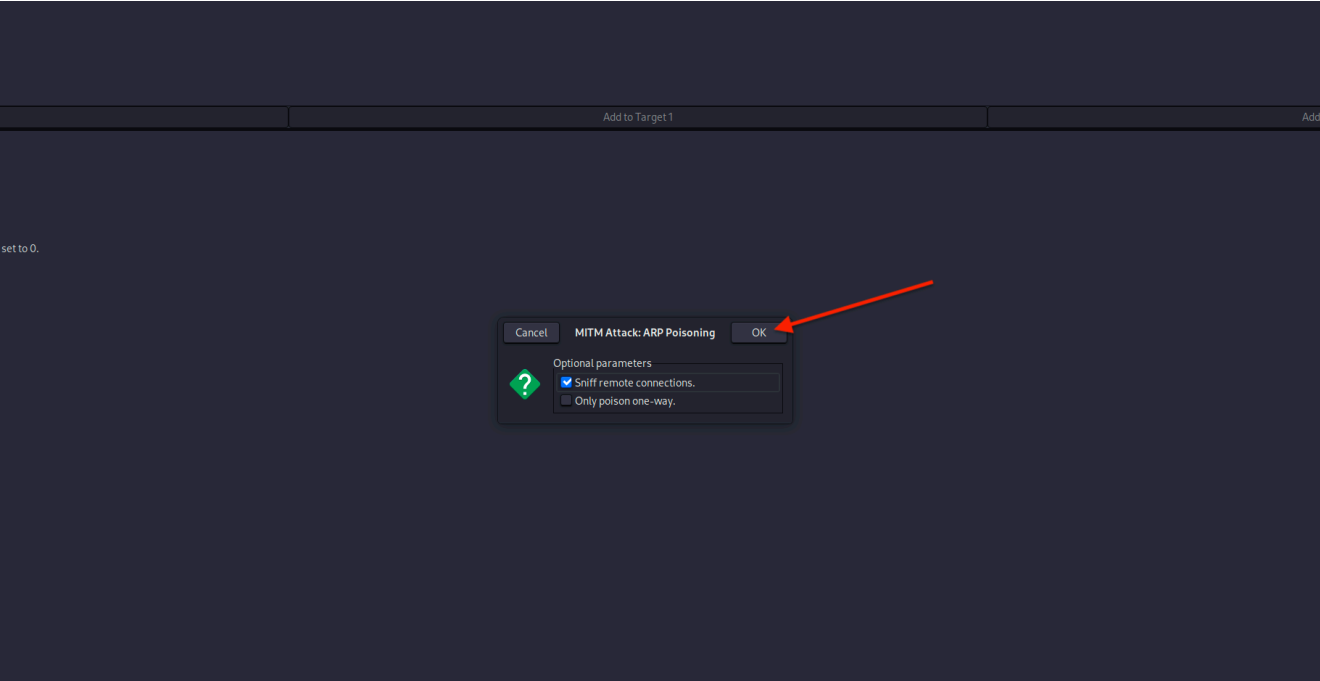
STEP 10 : Click on the globe icon to start the Man in The Middle attack



STEP 11 : In the drop down menu select ARP poisoning



Click on OK.



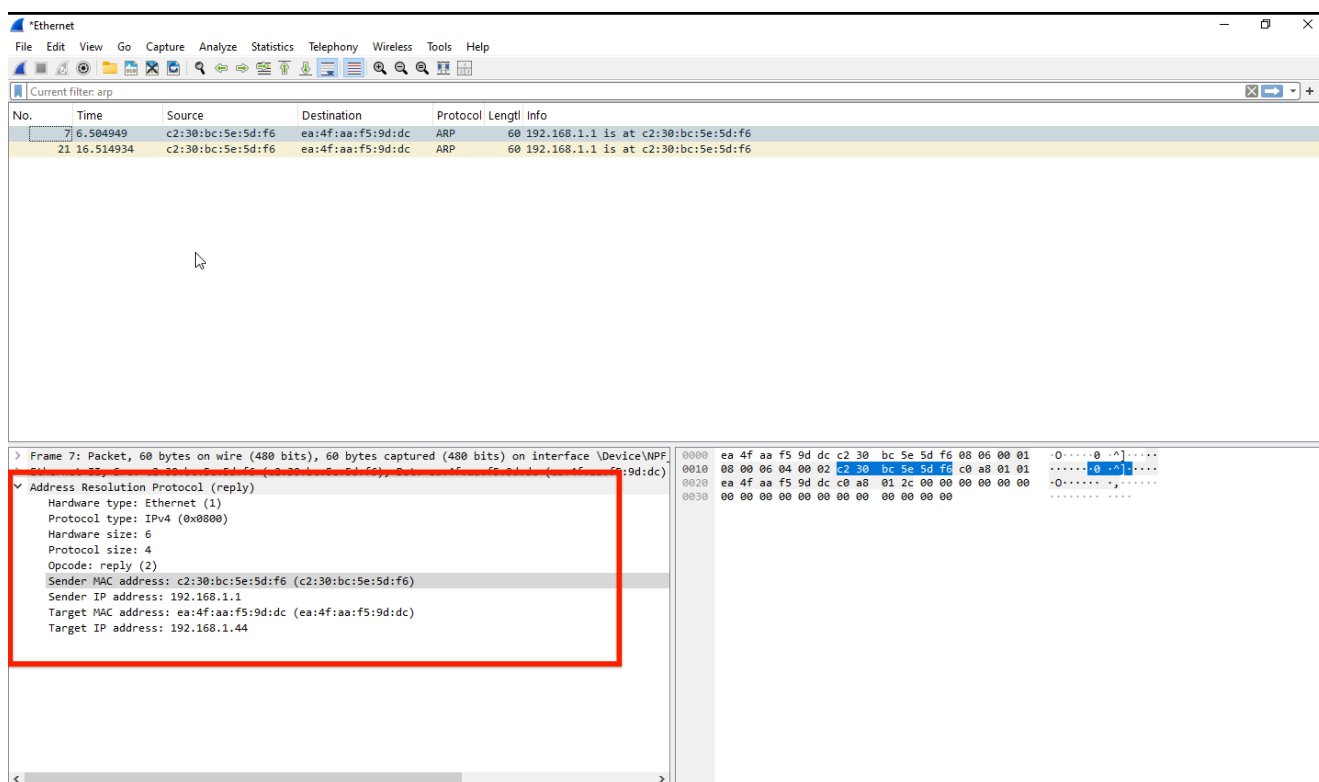
ARP Poisoning has been started



STEP 12 : Simultaneously start wireshark tool in the victim's machine and captured packets.

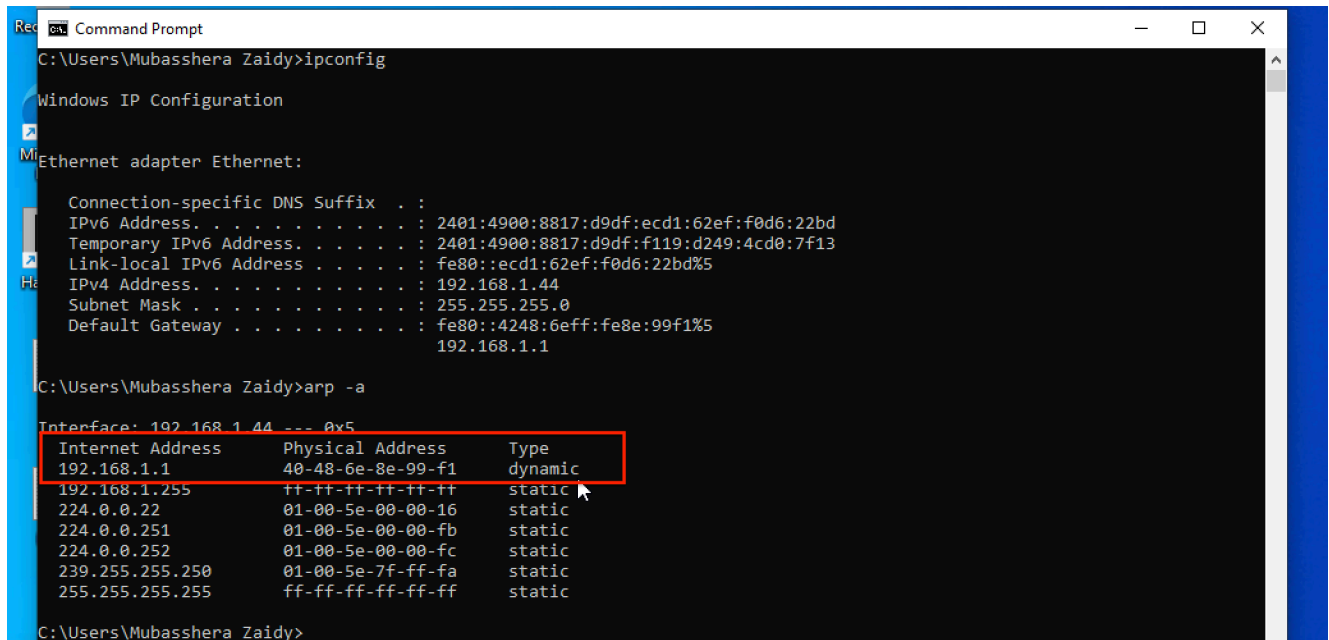
- Default Gateway : 192.168.1.1
- Attacker's MAC address : c2:30:bc:5e:5d:f6

Here under Address Resolution Protocol it's visible that the sender Ip Address is the default gateway - 192.168.1.1 but the sender's MAC address is the attacker's MAC address.



To see the effect this is having, open cmd and type arp -a
You can see that the arp cache has been changed and now the attacker's MAC address has replaced the default gateways MAC address

Before poisoning :



```

C:\Users\Mubasshera Zaidy>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

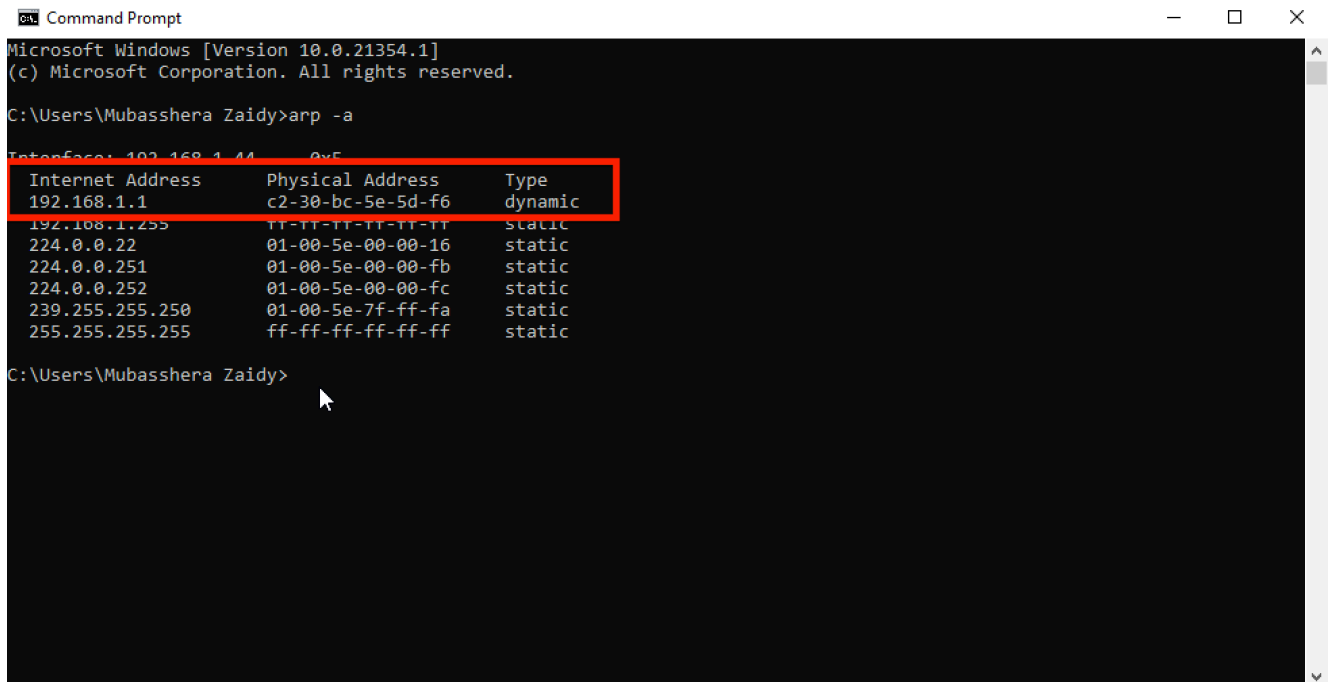
    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2401:4900:8817:d9df:ecd1:62ef:f0d6:22bd
    Temporary IPv6 Address. . . . . : 2401:4900:8817:d9df:f119:d249:4cd0:7f13
    Link-local IPv6 Address . . . . . : fe80::ecd1:62ef:f0d6:22bd%5
    IPv4 Address. . . . . : 192.168.1.44
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::4248:6eff:fe8e:99f1%5
                               192.168.1.1

C:\Users\Mubasshera Zaidy>arp -a

Interface: 192.168.1.44 --- 0x5
Internet Address      Physical Address      Type
192.168.1.1           40-48-6e-8e-99-f1    dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251          01-00-5e-00-00-fb    static
224.0.0.252          01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

```

After Poisoning :



```

Microsoft Windows [Version 10.0.21354.1]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mubasshera Zaidy>arp -a

Interface: 192.168.1.44 --- 0x5
Internet Address      Physical Address      Type
192.168.1.1           c2-30-bc-5e-5d-f6    dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251          01-00-5e-00-00-fb    static
224.0.0.252          01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

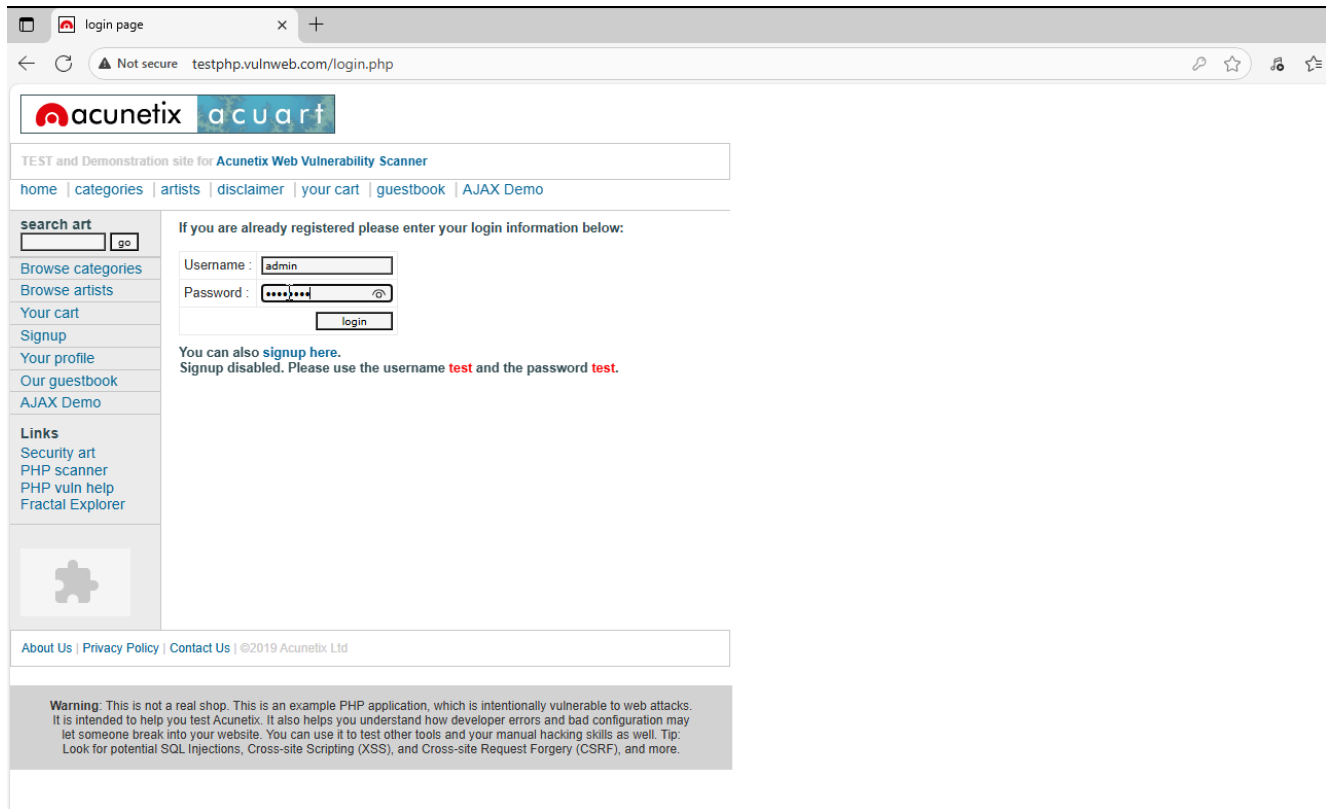
```

MAN-IN-THE-MIDDLE ATTACK

STEP 1 : In attacker's machine start wireshark tool.

No.	Time	Source	Destination	Protocol	Length	Info	
10.000	192.168.1.1	239.255.0.1	SS..	1..M-SEARCH	* HTTP/1.1		
20.204	192.168.1.1	239.255.0.1	SS..	1..M-SEARCH	* HTTP/1.1		
30.409	192.168.1.1	239.255.0.1	SS..	1..M-SEARCH	* HTTP/1.1		
40.759	c2:30:bc:ea:4f:aa	NokiaSol..	ARP	42	192.168.1.44 is at c2:30:bc:5e:5d:f6		
50.759	c2:30:bc:ea:4f:aa	ARP	42	192.168.1.1 is at c2:30:bc:5e:5d:f6 (duplicate use of 192.168.1.44 detected!)			
61.641	2401:4900:8817:d9df:d5fc:1c8d:7d24:89ea	IC..	86	Neighbor Solicitation for 2401:4900:8817:d9df:d5fc:1c8d:7d24:89ea from 40:48:6e:8e:99:f1			
71.641	2401:4900:8817:d9df:d5fc:1c8d:7d24:89ea	IC..	78	Neighbor Advertisement 2401:4900:8817:d9df:d5fc:1c8d:7d24:89ea (sol)			
82.559	NokiaSol..	IEEE-190..	ie..	60	Topology discovery		
94.601	fe80::c030:bfff:fe5e:5df6	IC..	86	Neighbor Solicitation for fe80::c030:bfff:fe5e:5df6 from 40:48:6e:8e:99:f1			
104.601	fe80::c030:bfff:fe5e:5df6	IC..	78	Neighbor Advertisement fe80::c030:bfff:fe5e:5df6 (sol)			
116.860	NokiaSol..	IEEE-190..	ie..	67	Vendor specific		
127.309	192.168.1.1	18.67.19..	TCP	55	50052 → 443 [ACK] Seq=1 Ack=1 Win=1021 Len=1		
137.309	192.168.1.1	18.67.19..	TCP	55	[TCP Keep-Alive] 50052 → 443 [ACK] Seq=1 Ack=1 Win=1021 Len=1		
147.329	18.67.19..	192.168.1.1	TCP	66	443 → 50052 [ACK] Seq=1 Ack=2 Win=140 Len=0 SLE=1 SRE=2		
157.329	18.67.19..	192.168.1.1	TCP	66	[TCP Keep-Alive ACK] 443 → 50052 [ACK] Seq=1 Ack=2 Win=140 Len=0 SLE=1 SRE=2		
167.616	2401:4900:8817:d9df:d5fc:1c8d:7d24:89ea	IC..	86	Neighbor Solicitation for 2401:4900:8817:d9df:d5fc:1c8d:7d24:89ea from 40:48:6e:8e:99:f1			
177.616	2401:4900:8817:d9df:d5fc:1c8d:7d24:89ea	IC..	78	Neighbor Advertisement 2401:4900:8817:d9df:d5fc:1c8d:7d24:89ea (sol)			
187.7963	192.168.1.1	18.67.19..	TCP	55	50055 → 443 [ACK] Seq=1 Ack=1 Win=1025 Len=1		
197.7963	192.168.1.1	18.67.19..	TCP	55	[TCP Keep-Alive] 50055 → 443 [ACK] Seq=1 Ack=1 Win=1025 Len=1		
207.7994	18.67.19..	192.168.1.1	TCP	66	443 → 50055 [ACK] Seq=1 Ack=2 Win=142 Len=0 SLE=1 SRE=2		
217.7994	18.67.19..	192.168.1.1	TCP	66	[TCP Keep-Alive ACK] 443 → 50055 [ACK] Seq=1 Ack=2 Win=142 Len=0 SLE=1 SRE=2		
227.8.159	192.168.1.1	20.189.1..	TCP	1..	50063 → 443 [ACK] Seq=1 Ack=1 Win=1022 Len=1250 [TCP PDU reassembled in 25]		
237.8.159	192.168.1.1	20.189.1..	TCP	1..	50063 → 443 [ACK] Seq=1251 Ack=1 Win=1022 Len=1250 [TCP PDU reassembled in 25]		
247.8.159	192.168.1.1	20.189.1..	TCP	1..	50063 → 443 [ACK] Seq=2501 Ack=1 Win=1022 Len=1250 [TCP PDU reassembled in 25]		
257.8.159	192.168.1.1	20.189.1..	TL..	4..	Application Data		
267.8.159	192.168.1.1	20.189.1..	TL..	93	Application Data		
277.8.159	192.168.1.1	20.189.1..	TCP	1..	[TCP Retransmission] 50063 → 443 [ACK] Seq=1 Ack=1 Win=1022 Len=1250		
287.8.159	192.168.1.1	20.189.1..	TCP	1..	[TCP Retransmission] 50063 → 443 [ACK] Seq=1251 Ack=1 Win=1022 Len=1250 [TCP PDU reassembled in 25]		
297.8.159	192.168.1.1	20.189.1..	TCP	1..	[TCP Retransmission] 50063 → 443 [ACK] Seq=2501 Ack=1 Win=1022 Len=1250 [TCP PDU reassembled in 25]		
307.8.159	192.168.1.1	20.189.1..	TCP	4..	[TCP Retransmission] 50063 → 443 [PSH, ACK] Seq=3751 Ack=1 Win=1022 Len=435 [TCP PDU reassembled in 25]		
317.8.159	192.168.1.1	20.189.1..	TCP	93	[TCP Retransmission] 50063 → 443 [PSH, ACK] Seq=4186 Ack=1 Win=1022 Len=39		
327.8.159	192.168.1.1	20.189.1..	TCP	1..	50063 → 443 [ACK] Seq=4225 Ack=1 Win=1022 Len=1250 [TCP PDU reassembled in 37]		
337.8.159	192.168.1.1	20.189.1..	TCP	1..	50063 → 443 [ACK] Seq=5475 Ack=1 Win=1022 Len=1250 [TCP PDU reassembled in 37]		
347.8.159	192.168.1.1	20.189.1..	TCP	1..	50063 → 443 [ACK] Seq=6725 Ack=1 Win=1022 Len=1250 [TCP PDU reassembled in 37]		
357.8.159	192.168.1.1	20.189.1..	TCP	1..	50063 → 443 [ACK] Seq=7975 Ack=1 Win=1022 Len=1250 [TCP PDU reassembled in 37]		
367.8.159	192.168.1.1	20.189.1..	TCP	1..	50063 → 443 [ACK] Seq=9225 Ack=1 Win=1022 Len=1250 [TCP PDU reassembled in 37]		

STEP 2 : As victim go to a http website which transmits data in plain text and enter your credentials ans click on login.



The screenshot shows a web browser window with the address bar displaying "testphp.vulnweb.com/login.php". The page features the Acunetix logo and a navigation menu with links like "home", "categories", "artists", "disclaimer", "your cart", "guestbook", and "AJAX Demo". A search bar is present on the left. The main content area contains a login form with fields for "Username" (containing "admin") and "Password" (masked with dots). A "login" button is below the password field. A message states: "If you are already registered please enter your login information below:". Below the login form, it says "You can also [signup here](#). Signup disabled. Please use the username **test** and the password **test**." The footer includes "About Us", "Privacy Policy", "Contact Us", and "©2019 Acunetix Ltd". A warning message at the bottom states: "Warning: This is not a real shop. This is an example PHP application, which is intentionally vulnerable to web attacks. It is intended to help you test Acunetix. It also helps you understand how developer errors and bad configuration may let someone break into your website. You can use it to test other tools and your manual hacking skills as well. Tip: Look for potential SQL Injections, Cross-site Scripting (XSS), and Cross-site Request Forgery (CSRF), and more."

login page

Not secure testphp.vulnweb.com/login.php

acunetix acuart

TEST and Demonstration site for [Acunetix Web Vulnerability Scanner](#)

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[Security art](#)

[PHP scanner](#)

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[Fractal Explorer](#)

If you are already registered please enter your login information below:

Username :

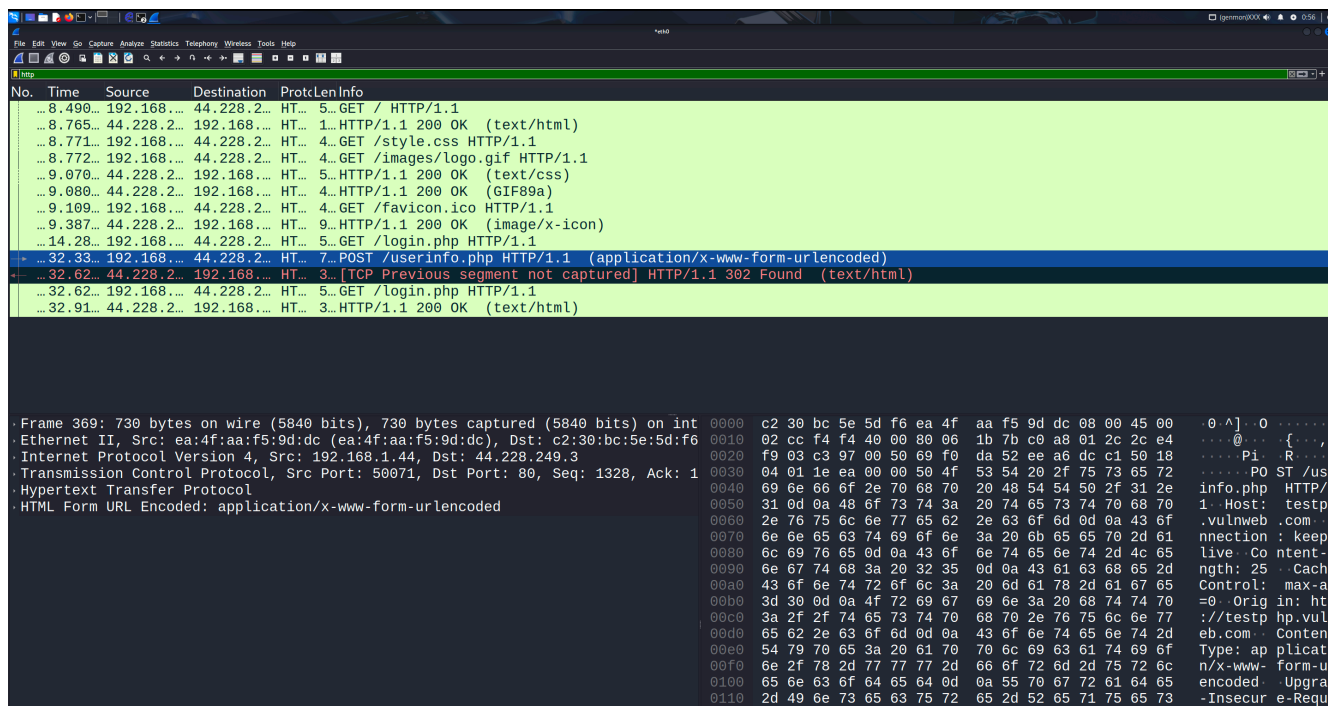
Password :

You can also [signup here](#).
Signup disabled. Please use the username **test** and the password **test**.

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Warning: This is not a real shop. This is an example PHP application, which is intentionally vulnerable to web attacks. It is intended to help you test Acunetix. It also helps you understand how developer errors and bad configuration may let someone break into your website. You can use it to test other tools and your manual hacking skills as well. Tip: Look for potential SQL Injections, Cross-site Scripting (XSS), and Cross-site Request Forgery (CSRF), and more.

STEP 3 : Now Come back to the attackers machine and stop packet capturing. In filter enter http to get the packets captured from http

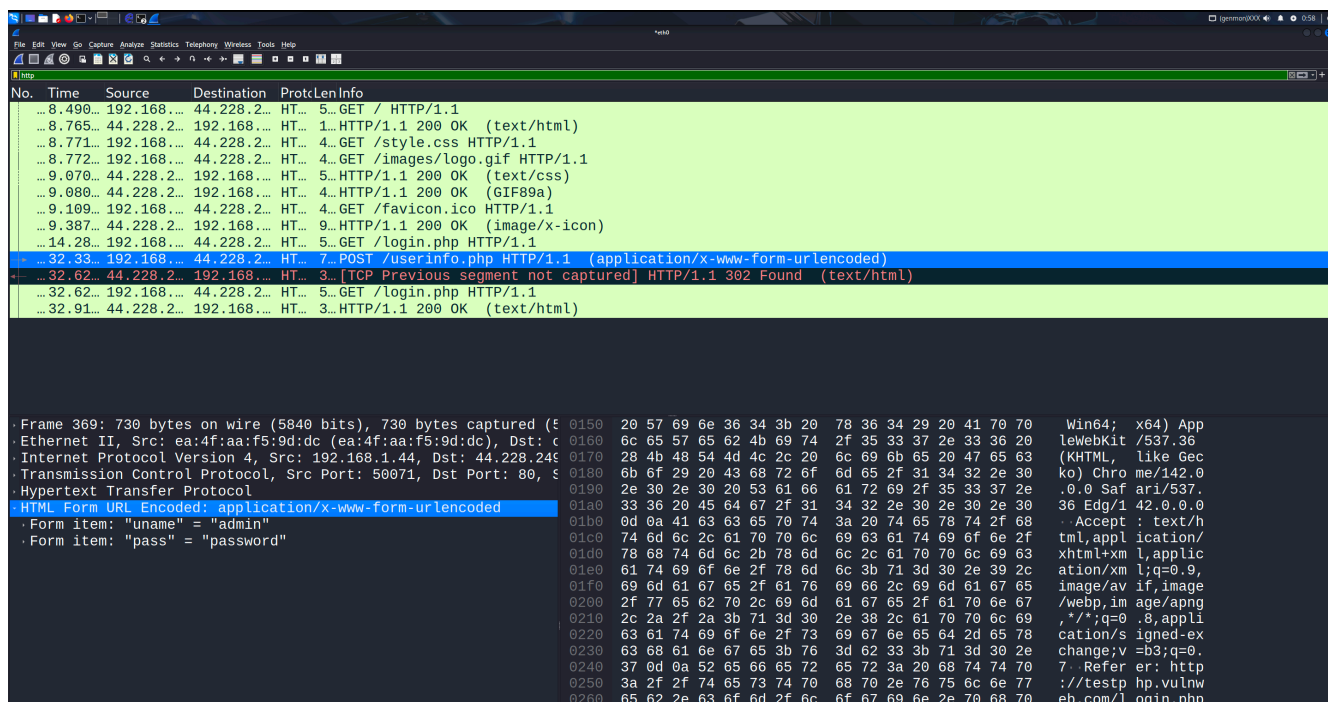


The image shows a Wireshark packet capture interface. The top pane displays a list of captured packets filtered by 'http'. The selected packet (No. 32.33) is a POST request to /userinfo.php. The bottom pane shows the packet details, including the Ethernet II header, Internet Protocol Version 4 header, Transmission Control Protocol header, and Hypertext Transfer Protocol header. The HTML Form URL Encoded data is visible in the bottom pane.

```
No. Time Source Destination ProtLenInfo
... 8.490... 192.168... 44.228.2... HT... 5..GET / HTTP/1.1
... 8.765... 44.228.2... 192.168... HT... 1..HTTP/1.1 200 OK (text/html)
... 8.771... 192.168... 44.228.2... HT... 4..GET /style.css HTTP/1.1
... 8.772... 192.168... 44.228.2... HT... 4..GET /images/logo.gif HTTP/1.1
... 9.070... 44.228.2... 192.168... HT... 5..HTTP/1.1 200 OK (text/css)
... 9.080... 44.228.2... 192.168... HT... 4..HTTP/1.1 200 OK (GIF89a)
... 9.109... 192.168... 44.228.2... HT... 4..GET /favicon.ico HTTP/1.1
... 9.387... 44.228.2... 192.168... HT... 9..HTTP/1.1 200 OK (image/x-icon)
... 14.28... 192.168... 44.228.2... HT... 5..GET /login.php HTTP/1.1
... 32.33... 192.168... 44.228.2... HT... 7..POST /userinfo.php HTTP/1.1 (application/x-www-form-urlencoded)
... 32.62... 44.228.2... 192.168... HT... 3..[TCP Previous segment not captured] HTTP/1.1 302 Found (text/html)
... 32.62... 192.168... 44.228.2... HT... 5..GET /login.php HTTP/1.1
... 32.91... 44.228.2... 192.168... HT... 3..HTTP/1.1 200 OK (text/html)

Frame 369: 730 bytes on wire (5840 bits), 730 bytes captured (5840 bits) on int
Ethernet II, Src: ea:4f:aa:f5:9d:dc (ea:4f:aa:f5:9d:dc), Dst: c2:30:bc:5e:5d:f6
Internet Protocol Version 4, Src: 192.168.1.44, Dst: 44.228.249.3
Transmission Control Protocol, Src Port: 50071, Dst Port: 80, Seq: 1328, Ack: 1
Hypertext Transfer Protocol
HTML Form URL Encoded: application/x-www-form-urlencoded
```

Under the POST packet you'll see that the data which the victim machine had entered is intercepted by the attacker's machine.



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Transmission Control Protocol, Src Port: 50071, Dst Port: 80, Seq: 1328, Ack: 1
Hypertext Transfer Protocol
HTML Form URL Encoded: application/x-www-form-urlencoded
Form item: "uname" = "admin"
Form item: "pass" = "password"
```

Result

This result confirms that ARP cache poisoning successfully redirected traffic through the attacker, enabling a **man-in-the-middle attack**. The victim's internet connectivity remained intact due to IP forwarding, while the attacker gained visibility into the victim's communications.